

CS 171 / EE 142: Intro to ML and DM

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UC Riverside

Slide Set 7: Gradient Descent



- From UC Riverside
 - ▶ CS 171 / EE 142: Introduction to Machine Learning and Data Mining
 - ▶ Professor Christian Shelton
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 - ▶ For use only by enrolled students in the course

Gradient Descent

What if you have a function you want to minimize...

Gradient Descent

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...But! setting the derivatives equal to zero gives equations you cannot solve (analytically)?

Gradient Descent

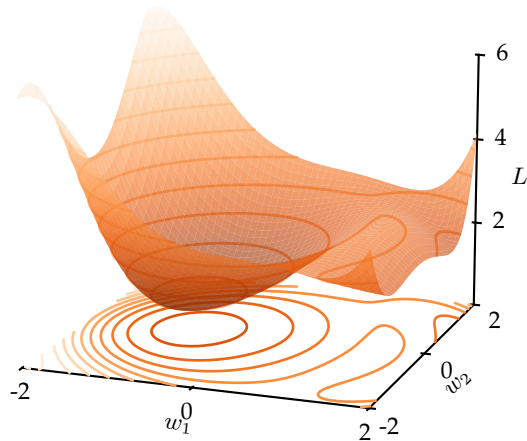
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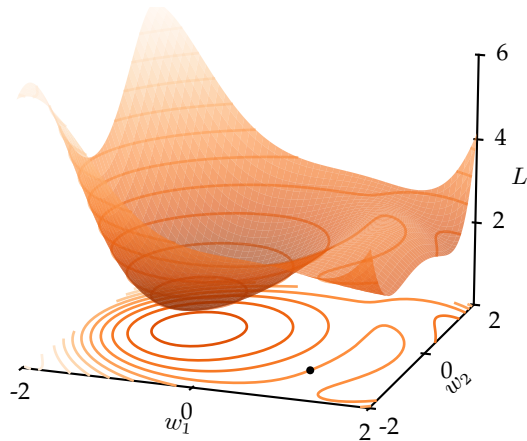
Numeric optimization!

The workhorse of numeric optimization: **gradient descent** (GD)

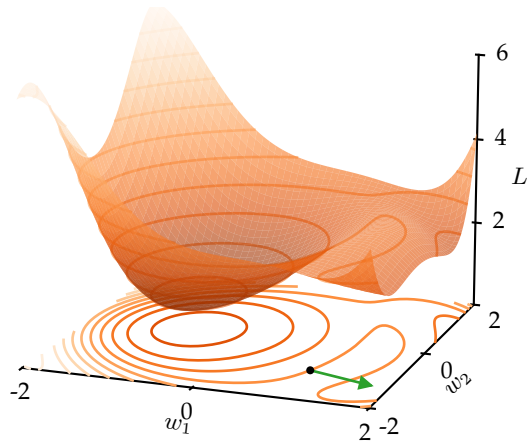
Perceptron Learning Algorithm, Example



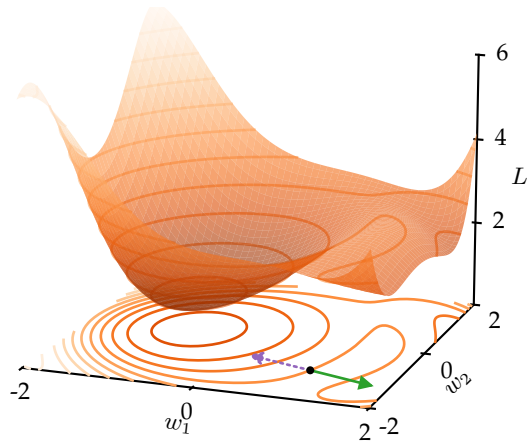
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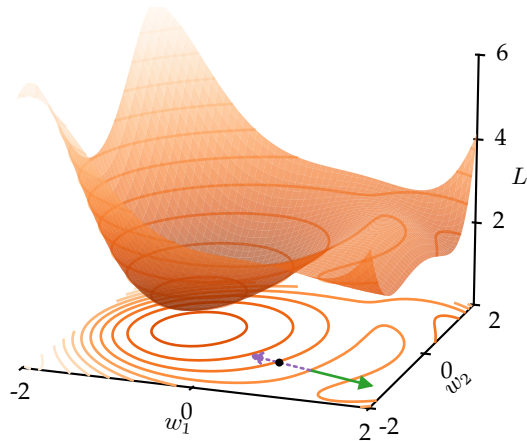
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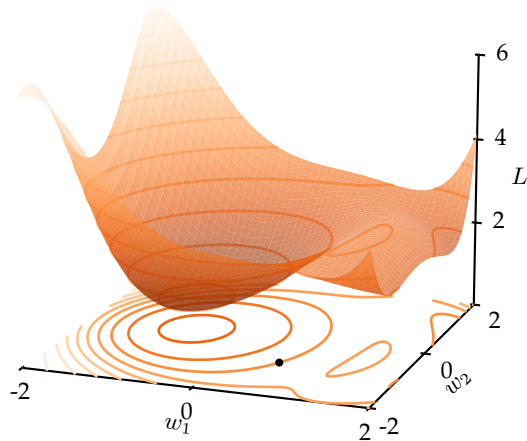
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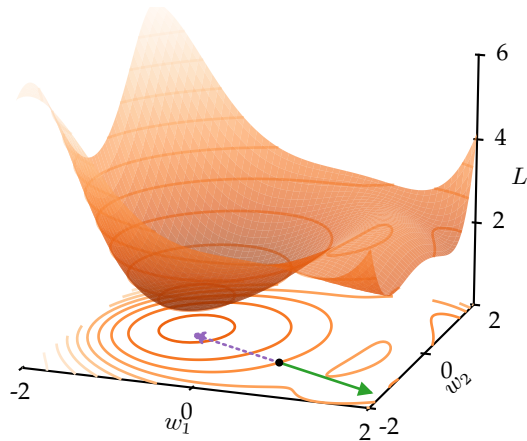
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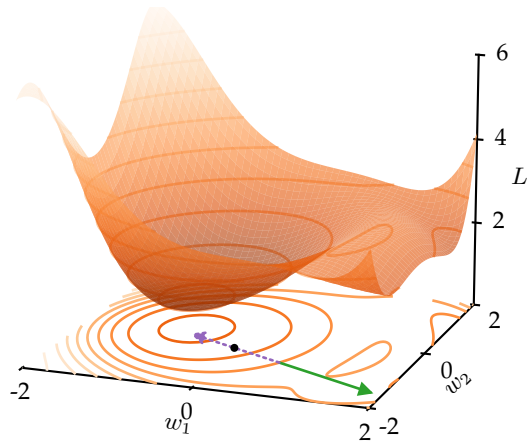
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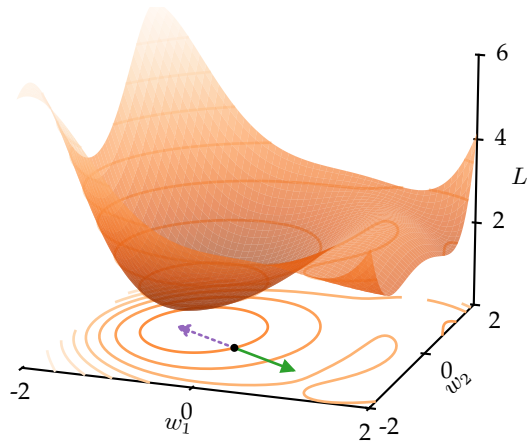
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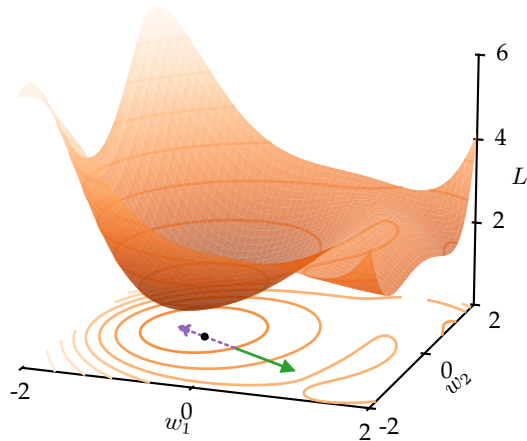
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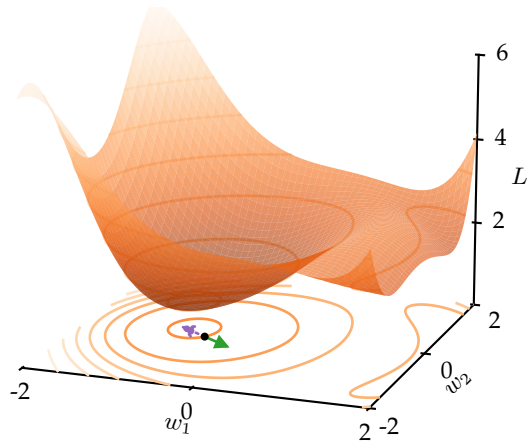
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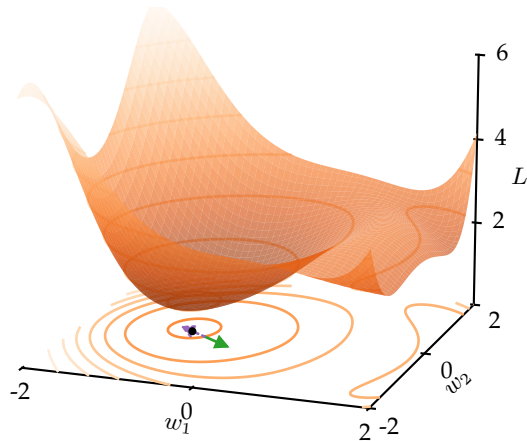
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Perceptron Learning Algorithm, Example



Gradient Descent Algorithm

- 1 Let $\vec{w}$ be a random point
- 2 While $\vec{w}$ is not at local minimum of L
 - 1 Let $\vec{g} \leftarrow \nabla_{\vec{w}} L(\vec{w})$
 - 2 Let $\vec{w} \leftarrow \vec{w} - \eta \vec{g}$

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Clarifying points:

- **Step size (η):**
 - ▶ Too small: many many steps
 - ▶ Too large: jump over minima and may diverge

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- **Convergence**
 - ▶ Option 1: stop when gradient ($\nabla_{\vec{w}} L$) is small
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Clarifying points:

- **Step size (η):**
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 - ▶ In practice: use *adaptive step size*
- **Convergence**
 - ▶ Option 1: stop when gradient ($\nabla_{\vec{w}} L$) is small
 - ▶ Option 2: stop when objective (L) stops changing much
- **Starting point**
 - ▶ Ideally, near the minimum
 - ▶ Often, should avoid $\vec{w} = \vec{0}$ due to symmetries

Stochastic Gradient Descent

If

$$L(\vec{w}) = \sum_{i=1}^m l_i(\vec{w})$$

Then

$$\nabla_{\vec{w}} L(\vec{w}) = \sum_{i=1}^m \nabla_{\vec{w}} l_i(\vec{w})$$

Stochastic Gradient Descent

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Gradient Descent (GD)

- ① Let $\vec{w}$ be a random point
- ② While $\vec{w}$ is not at local minimum of L
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Gradient Descent (GD)

- ① Let $\vec{w}$ be a random point
- ② While $\vec{w}$ is not at local minimum of L
 - ① Let $\vec{g} \leftarrow 0$
 - ② For $i = 1, \dots, m$
 - ① Let $\vec{g}_i \leftarrow \nabla_{\vec{w}} l_i(\vec{w})$
 - ② Let $\vec{g} \leftarrow \vec{g} + \vec{g}_i$
 - ③ Let $\vec{w} \leftarrow \vec{w} - \eta \vec{g}$

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- Only applicable when L has this sum form.
- Instead of waiting to update $\vec{w}$ until the gradient has been completely summed, update after each component is computed
- Tends to bounce around more, but can converge faster in many cases
- Randomization of order of points is important